

GRAIN - Grocery product Recognition and Analysis in Unconstrained Environments

1 DATASET CHARACTERISTICS

GRAIN (Grocery product Recognition and Analysis in Unconstrained Environments) is a multi-task dataset designed to support product detection, label localization, and text recognition in real-world home settings. It includes over 5,700 high-resolution images captured using mobile devices under varying lighting conditions, simulating natural household use-cases including handheld and obstructed views. The dataset is divided into three annotated and partially overlapping subsets: (1) annotations for detecting a set of common product categories (2) region annotations for label detection; and (3) semantic-level annotations for textual fields on labels.

The dataset is divided into three fully annotated and partially overlapping subsets:

- **Product detection annotations** across 5 categories: wine bottles, water bottles, soft drinks, olive oil, and vinegar (1,750 images, 4,388 annotations)
- **Wine label detection annotations**, accounting for curvature and occlusion (2,463 images, 4,669 annotations)
- **Semantic-level text annotations** for labels, including brand, wine type, region, vintage year, alcohol percentage, volume, and country (1,579 images, 7,610 annotations). A subset of these annotations includes transcriptions for OCR benchmarking.

All images were acquired in diverse home settings and a small subset in supermarkets, aiming to provide robustness against varied backgrounds, lighting, occlusion, and camera angles (Figure 1).

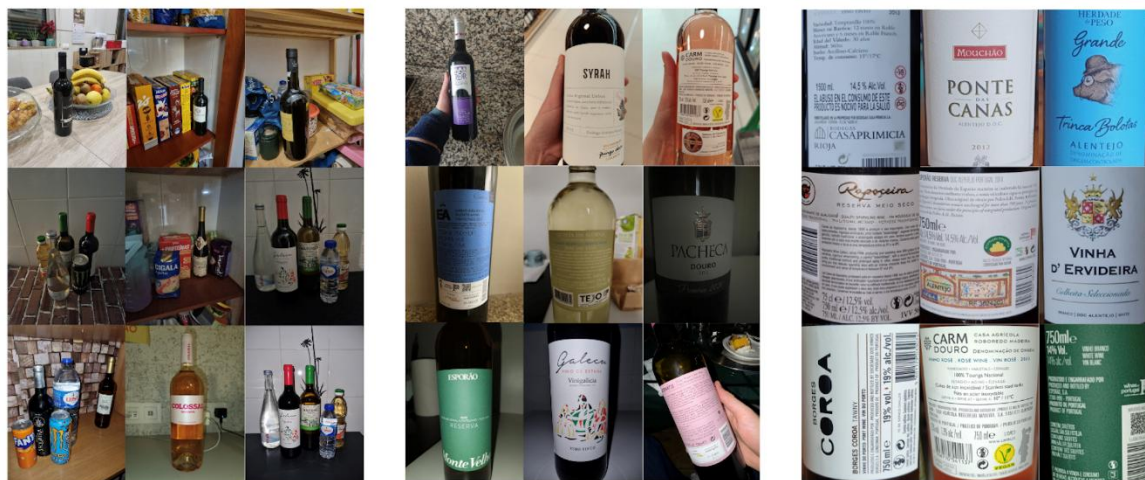


Figure 1: Sample images from the GRAIN dataset, illustrating examples used across the three different subsets.

2 IMAGE ACQUISITION PROTOCOL

Images were captured with mobile phone cameras ($\sim 3000 \times 4000$ px) using the smartphones' native apps, with automatic settings and no filters. Scenes were designed to reflect:

- **Lighting conditions:** good (daylight or overhead), low-light, and flash.
- **Scenes:** simple (single product), composite (multiple overlapping products), and obstructed (partially hidden products).
- **Conditions:** multiple users captured photos mostly in home settings, interacting naturally with the products.

3 ANNOTATION STRATEGY

Each subset is annotated according to task requirements:

- **Product detection:** bounding boxes per product class in COCO format.
- **Label localization:** bounding boxes around main wine labels in COCO format.
- **Text recognition:** bounding boxes for seven semantic fields in COCO format, with transcriptions for a subset.

The dataset is organized into a hierarchical folder structure that reflects both the task-specific annotation goals and the different capture conditions. At the top level, the dataset is divided into folders corresponding to the three main tasks.

Within each task-specific folder, images are further grouped according to lighting conditions.

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DISCLAIMER

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